

SMART CITY / MOBILITY ANALYTICS

Parking Demand Prediction

using Elastic Net Regularized Regression

Forecasting where the cars will be — before they get there.

MACHINE LEARNING MINI PROJECT

Guide: Shaik Asif

Assistant Professor, CSE

Nuguri Siri – 2520030309

S. Manvi Vaishnavi – 2520030326

1. Problem Statement

The lot is full. Nobody saw it coming.

Parking capacity stays **fixed** while vehicle demand changes by **hour, day, weather and events**.

TODAY'S REALITY

Drivers cruise blind

Operators react blind

LIVE OCCUPANCY: **FULL** → the sign tells you only when you arrive.

What a full lot costs:

1

TIME

Lost circling for a space

2

TRAFFIC

Congestion around full lots

3

FUEL

Fuel and emissions while idling

The goal: The sign tells you when you arrive. **Prediction** tells you before you leave.

2. Project Abstract

From raw records to an early forecast.

1 Objective

Forecast parking demand ahead of time so limited capacity can be managed by plan, not reaction.

2 Data + Method

Occupancy, time/date, weather, holidays and events → cleaning → feature engineering → Elastic Net.

3 Outcome

Forecasts by location and time window, evaluated with MAE, RMSE and R^2 .

PREDICT → **PLAN** → **ACT** Capacity managed by plan, not reaction.

3. Existing System

Why existing systems fall short.

1

Manual Logs

Paper registers

After-the-fact

2

Sensors + Displays

Bay sensors / cameras

Present moment only

3

Static Rules

Fixed allocation / pricing

No context

4

Simple Regression

Basic trend estimate

Weak with correlation

The gap: existing systems are **descriptive**. This project is **predictive**.

4. Proposed Solution

“One road. Five stages. One forecast.” Elastic Net regression turns parking history into an early demand forecast.



Output: Operators · planners · traffic authorities receive the forecast early enough to act.

5. Research Papers

6. Why Elastic Net

Two penalties. One model.

LASSO

L1 selects

Weak coefficients can shrink to zero → irrelevant features drop out.

RIDGE

L2 stabilises

Correlated predictors are handled smoothly → variance is controlled.



ELASTIC NET

Select + Stabilise

Parking features are often correlated:

hour

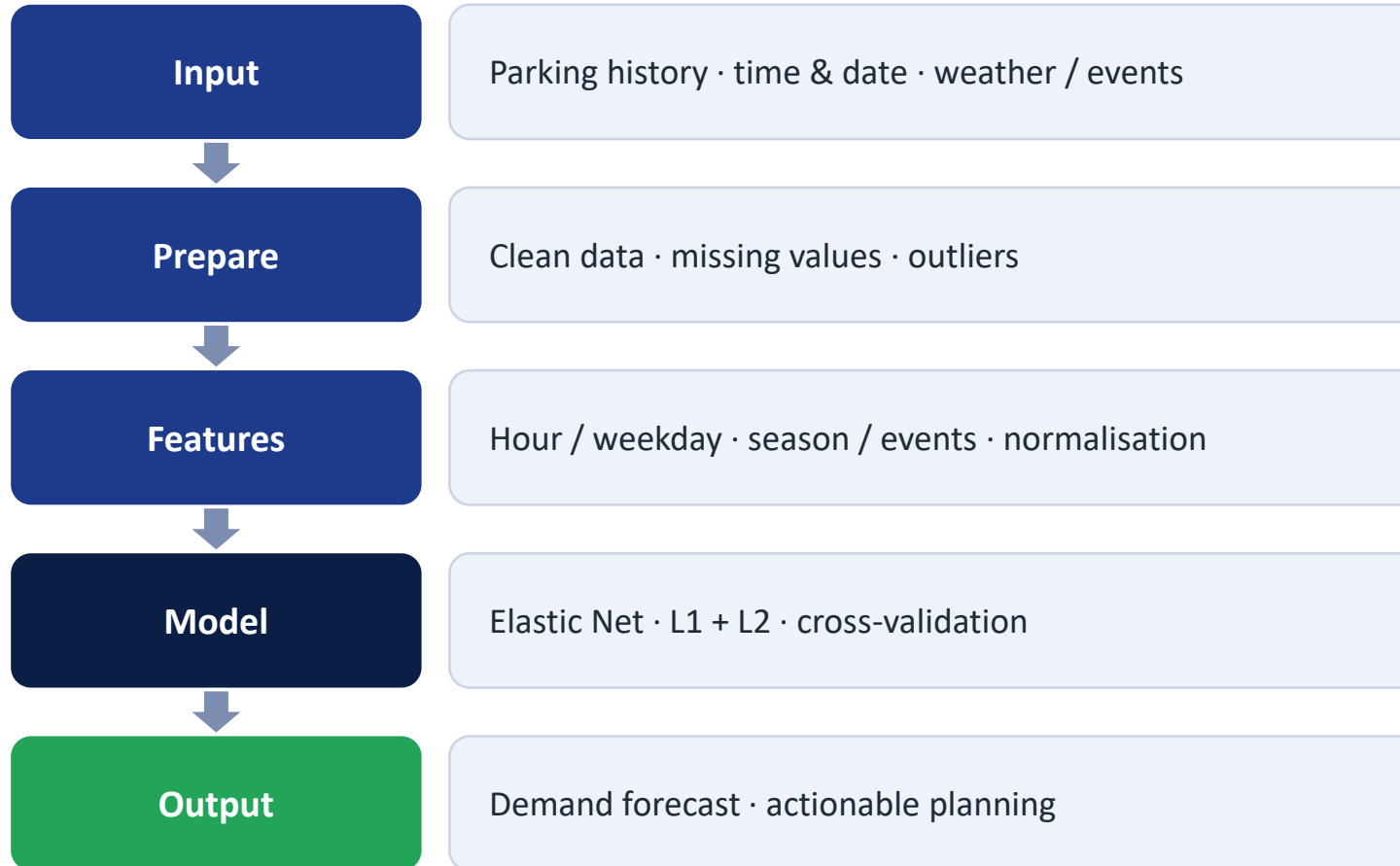
weekday

weather

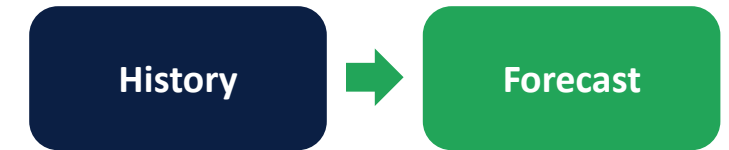
events

Tuning: the L1/L2 mixing ratio and penalty strength are tuned by **cross-validation**.

7. Model Workflow



In short



The system learns from past parking records and context, and turns them into a demand forecast that supports better parking decisions.

**History → Features → Elastic Net → Forecast →
Better parking decisions**

8. Applications & Future Scope

Where the forecast can make a difference

1 Driver Guidance

1

Show expected availability before arrival and reduce time spent searching.

2 Lot Operations

2

Plan bay allocation and respond to changing demand by time window.

3 Traffic Planning

3

Use demand forecasts to anticipate congestion around busy parking areas.

4 Smarter Policy

4

Support evidence-based allocation, pricing and capacity planning.

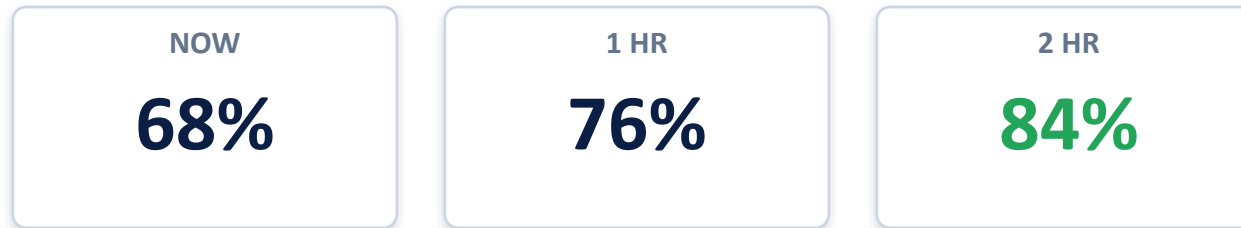
Next step: connect forecasts to real parking availability and operational decisions.

Thank You

Questions?

Smarter parking, smoother cities.

PREDICTED OCCUPANCY • DEMAND TREND



Elastic Net regularized regression turns parking history into foresight — and foresight into better urban mobility.

Nuguri Siri · S. Manvi Vaishnavi | Guide: Shaik Asif, Assistant Professor, CSE